

Data Validation Query 1 — Row Count and Null Check

Business Question: How many rows are in the clean dataset? Are there any missing values in key columns?

	total_rows bigint	invoice_no_count bigint	customer_id_count bigint	description_count bigint	missing_customer_ids bigint
1	530100	530100	397880	530100	132220

Key Findings

- The clean table has 530,100 rows after removing returns, pricing errors, and blank descriptions.
- All invoice numbers and descriptions are present with no nulls in those columns.
- 132,220 rows have no Customer ID — roughly 25% of transactions are from unidentified buyers, so customer-level analysis covers only 397,880 rows.

Data Validation Query 2 — Range Check

Business Question: Are there any unexpected minimum or maximum values in quantity, price, or revenue after cleaning?

	min_quantity integer	max_quantity integer	min_unit_price numeric	max_unit_price numeric	min_revenue numeric	max_revenue numeric	earliest_date timestamp without time zone	latest_date timestamp without time
1	1	80995	0.04	13541.33	0.06	168469.60	2010-12-01 08:26:00	2011-12-09 12:50:00

Key Findings

- Minimum quantity is 1 and minimum unit price is £0.04 — cleaning worked correctly, no negatives slipped through.
- Maximum quantity is 80,995 and maximum unit price is £13,541.33 — these are extreme outliers worth noting in analysis.
- Dataset spans December 2010 to December 2011, giving roughly 13 months of transaction data.

Query 1 — Revenue by Country (Excluding UK)

Business Question: Which international markets generate the most revenue, and how does quantity compare across them?

	country text	total_revenue numeric	total_quantity bigint	total_orders bigint
1	Netherlands	285446.34	200361	94
2	EIRE	283453.96	147173	288
3	Germany	228867.14	119261	457
4	France	209715.11	112103	392
5	Australia	138521.31	83901	57
6	Spain	61577.11	27940	90
7	Switzerland	57089.90	30629	54
8	Belgium	41196.34	23237	98
9	Sweden	38378.33	36083	36
10	Japan	37416.37	26016	19

Key Findings

- Netherlands leads international revenue at £285,446 across just 94 orders — very high average order value, suggesting bulk wholesale buyers.
- EIRE (Ireland) is close behind at £283,454 with 288 orders — more frequent but smaller purchases compared to Netherlands.
- Top 5 markets (Netherlands, EIRE, Germany, France, Australia) account for the majority of international revenue and are the priority expansion markets.

Query 2 — Monthly Revenue Trend 2011

Business Question: How did revenue trend month by month in 2011? Are there seasonal peaks?

	month_number numeric	month_name text	total_revenue numeric	total_orders bigint
1	1	January	691364.56	1086
2	2	February	523631.89	1100
3	3	March	717639.36	1454
4	4	April	537808.62	1246
5	5	May	770536.02	1681
6	6	June	761739.90	1533
7	7	July	719221.19	1475
8	8	August	759138.38	1361
9	9	September	1058590.17	1837
10	10	October	1154979.30	2040
11	11	November	1509496.33	2769
12	12	December	638792.68	819

Key Findings

- November is the strongest month at £1,509,496 — driven by early Christmas season wholesale orders.
- September and October also show strong performance at £1,058,590 and £1,154,979 — Q4 build-up is clear.
- December drops to £638,793 because the dataset ends on 9 December — this is incomplete data, not a real seasonal decline.

Query 3 — Top 10 Customers by Revenue

Business Question: Which customers generate the most revenue, and how concentrated is that value?

	customer_id text	total_revenue numeric	total_orders bigint	total_quantity bigint
1	14646	280206.02	73	196915
2	18102	259657.30	60	64124
3	17450	194550.79	46	69993
4	16446	168472.50	2	80997
5	14911	143825.06	201	80265
6	12415	124914.53	21	77374
7	14156	117379.63	55	57885
8	17511	91062.38	31	64549
9	16029	81024.84	63	40208
10	12346	77183.60	1	74215

Key Findings

- Customer 14646 is the top buyer at £280,206 across 73 orders — highest revenue with a moderate order count.
- Customer 16446 spent £168,473 in just 2 orders — extremely high-value single transactions, likely a bulk wholesaler.
- Top 10 customers represent heavy revenue concentration — losing even one of these accounts would significantly impact the business.

Query 4 — Top 10 Products by Revenue

Business Question: Which products drive the most revenue for the business?

	stock_code text	description text	total_revenue numeric	total_quantity bigint	total_orders bigint
1	DOT	DOTCOM POSTAGE	206248.77	706	706
2	22423	REGENCY CAKESTAND 3 TIER	174484.74	13879	1988
3	23843	PAPER CRAFT , LITTLE BIRDIE	168469.60	80995	1
4	85123A	WHITE HANGING HEART T-LIGHT HOLD...	104340.29	37599	2189
5	47566	PARTY BUNTING	99504.33	18295	1685
6	85099B	JUMBO BAG RED RETROSPOT	94340.05	48474	2089
7	23166	MEDIUM CERAMIC TOP STORAGE JAR	81700.92	78033	247
8	M	Manual	78110.27	7224	289
9	POST	POSTAGE	78101.88	3150	1126
10	23084	RABBIT NIGHT LIGHT	66964.99	30788	994

Key Findings

- DOTCOM POSTAGE ranks first at £206,249 — this is a shipping charge, not a real product, and should be excluded from product analysis.
- REGENCY CAKESTAND 3 TIER is the top real product at £174,485 across 1,988 orders — a genuine bestseller with strong volume.
- PAPER CRAFT LITTLE BIRDIE at £168,470 from just 1 order suggests a single massive bulk purchase — a data anomaly worth flagging.

Query 5 — Repeat Customer Analysis

Business Question: Which customers made the most purchases over the period, and how long have they been active?

	customer_id text	total_orders bigint	total_revenue numeric	first_purchase date	last_purchase date
1	12748	209	33719.73	2010-12-01	2011-12-09
2	14911	201	143825.06	2010-12-01	2011-12-08
3	17841	124	40991.57	2010-12-01	2011-12-08
4	13089	97	58825.83	2010-12-05	2011-12-07
5	14606	93	12156.65	2010-12-01	2011-12-08
6	15311	91	60767.90	2010-12-01	2011-12-09
7	12971	86	11189.91	2010-12-02	2011-12-06
8	14646	73	280206.02	2010-12-20	2011-12-08
9	16029	63	81024.84	2010-12-01	2011-11-01
10	13408	62	28117.04	2010-12-01	2011-12-08

Key Findings

- Customer 12748 placed 209 orders — the most frequent buyer, active from December 2010 to December 2011 across the full period.
- Most top repeat customers have been active since the very start of the dataset, showing strong long-term loyalty.
- High order count does not always mean high revenue — Customer 12971 has 86 orders but only £11,190 revenue versus Customer 14646's 73 orders and £280,206.

Query 6 — Average Order Value by Country (Excluding UK)

Business Question: Which markets have the highest value per order, not just total revenue?

	country text	total_orders bigint	total_revenue numeric	avg_order_value numeric
1	Singapore	7	21279.29	3039.90
2	Netherlands	94	285446.34	3036.66
3	Australia	57	138521.31	2430.20
4	Japan	19	37416.37	1969.28
5	Lebanon	1	1693.88	1693.88
6	Hong Kong	11	15691.80	1426.53
7	Brazil	1	1143.60	1143.60
8	Sweden	36	38378.33	1066.06
9	Switzerland	54	57089.90	1057.22
10	Denmark	18	18955.34	1053.07

Key Findings

- Singapore has the highest average order value at £3,040 per order despite only 7 total orders — very high-value but low-frequency market.
- Netherlands has the second highest AOV at £3,037 — confirming Dutch buyers are placing large wholesale orders rather than frequent small ones.
- Australia at £2,430 AOV with 57 orders shows both high value and decent frequency — a strong growth market worth prioritising.

Query 7 — Revenue by Quarter

Business Question: Which quarter drives the most revenue? Is there a strong seasonal pattern across the year?

	year numeric	quarter numeric	total_revenue numeric	total_orders bigint	total_quantity bigint
1	2010	4	823746.14	1559	359239
2	2011	1	1932635.81	3640	1048866
3	2011	2	2070084.54	4460	1093765
4	2011	3	2536949.74	4673	1394346
5	2011	4	3303268.31	5628	1692156

Key Findings

- Q4 2011 is the strongest quarter at £3,303,268 — confirming a heavy end-of-year seasonal peak driven by Christmas wholesale orders.
- Revenue grows consistently each quarter through 2011: Q1 £1.9M, Q2 £2.1M, Q3 £2.5M, Q4 £3.3M — a clear upward trend.
- Q4 2010 at £823,746 covers only December and is not comparable to full quarters — noted as a data limitation.

Query 8 — Customer Purchase Frequency Segments

Business Question: What percentage of customers are one-time buyers versus repeat buyers?

	customer_segment text	customer_count bigint	percentage numeric
1	1 - One Time Buyer	1493	34.42
2	11 Plus - Loyal Buyer	337	7.77
3	2 to 5 - Occasional Buyer	1973	45.48
4	6 to 10 - Regular Buyer	535	12.33

Key Findings

- 45.48% of customers are occasional buyers (2–5 orders) — the largest segment, showing most customers do return but infrequently.
- 34.42% are one-time buyers — over one third of the customer base never returned, which is a significant retention risk.
- Only 7.77% are loyal buyers (11+ orders) — a small but high-value segment likely driving a disproportionate share of total revenue.

Query 9 — Top 10 Countries by Quantity Demanded (Excluding UK)

Business Question: Which markets have the highest product demand by volume, not just revenue?

	country text	total_quantity bigint	total_revenue numeric	total_orders bigint	demand_rank bigint
1	Netherlands	200361	285446.34	94	1
2	EIRE	147173	283453.96	288	2
3	Germany	119261	228867.14	457	3
4	France	112103	209715.11	392	4
5	Australia	83901	138521.31	57	5
6	Sweden	36083	38378.33	36	6
7	Switzerland	30629	57089.90	54	7
8	Spain	27940	61577.11	90	8
9	Japan	26016	37416.37	19	9
10	Belgium	23237	41196.34	98	10

Key Findings

- Netherlands tops quantity demand at 200,361 units across only 94 orders — confirms these are large bulk wholesale orders.
- EIRE (Ireland) is second at 147,173 units — high in both revenue and volume, making it the most consistently strong international market.
- Sweden appears in the quantity top 10 at 36,083 units but ranked lower in revenue — lower-priced items being bought in volume.

Query 10 — Revenue Contribution with Running Total (Excluding UK)

Business Question: What percentage of international revenue does each country contribute, and what does the cumulative picture look like?

	country text	total_revenue numeric	revenue_percentage numeric	running_total numeric
1	Netherlands	285446.34	17.39	285446.34
2	EIRE	283453.96	17.27	568900.30
3	Germany	228867.14	13.94	797767.44
4	France	209715.11	12.78	1007482.55
5	Australia	138521.31	8.44	1146003.86
6	Spain	61577.11	3.75	1207580.97
7	Switzerland	57089.90	3.48	1264670.87
8	Belgium	41196.34	2.51	1305867.21
9	Sweden	38378.33	2.34	1344245.54
10	Japan	37416.37	2.28	1381661.91

Key Findings

- Netherlands and EIRE together contribute 34.66% of all international revenue — nearly a third from just two markets.
 - Top 5 countries (Netherlands, EIRE, Germany, France, Australia) account for 69.82% of international revenue cumulatively.
 - This heavy concentration means the business is dependent on a few markets — diversification into mid-tier countries like Spain and Switzerland should be explored.
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